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Efficient Joke Transformer

Initial Talk, Efficient Methods for ML

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Research Question

- Motivation
  - *Humor reflects intelligence and creativity — can small, efficient transformers learn it from scratch?*
  - *Modern generative LLMs (GPT, Llama, etc.) are decoder-only — why is this the trend, and how does it work?*

- Goal

  Explore efficient transformer architectures for joke generation, conditioned on user-provided topics/nouns.

Datasets

- Datasets
  1. Short Jokes ~230k
  2. rJokesData ~350k
  3. Amirkid Jokes ~580k
  4. ColBERT for Humor Detection 100k
  5. Dadjokes ~50k
- Preprocessing
  - Data cleaning & topic extraction
- Outcome
  - Cleaned dataset (~726k) ready for training and evaluation

Raw Data	Why did the **chicken** cross the **road**? To get to the other **side**!
Extracted Topics	chicken, road, side
Clean Data	{ tags: [chicken, road, side], joke: “Why did the chicken …” }

POS tagging

Two Architectures

Encoder–Decoder Transformer

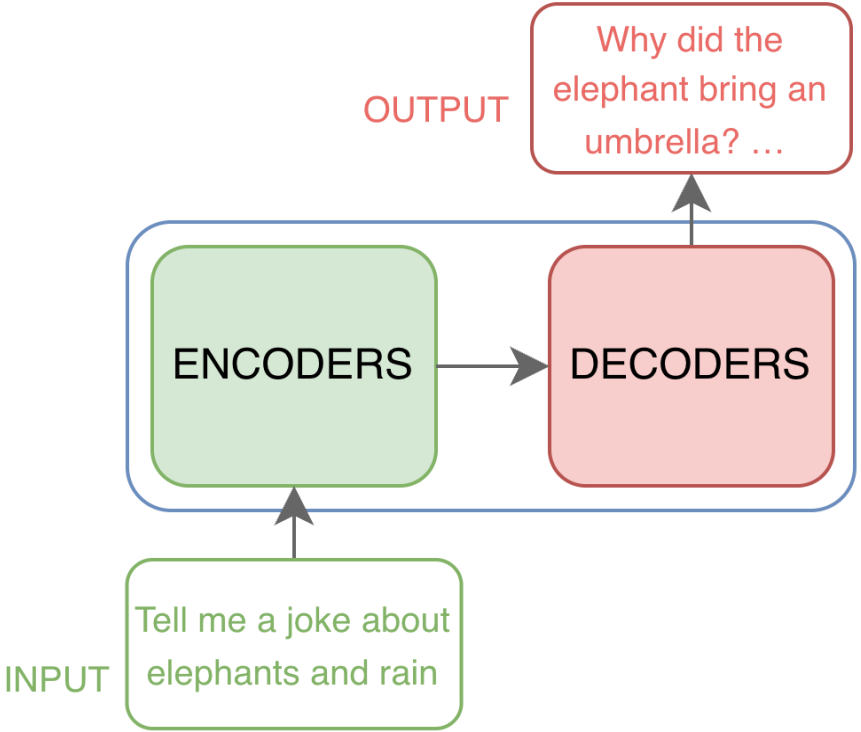
- Encoder input

  "Tell me a joke about elephants and rain"

- Decoder output

  "Why did the elephant bring an umbrella? …"

- Expected features
  - Clear conditioning boundary
  - More computationally expensive



Two Architectures

Decoder-Only Transformer

- Input

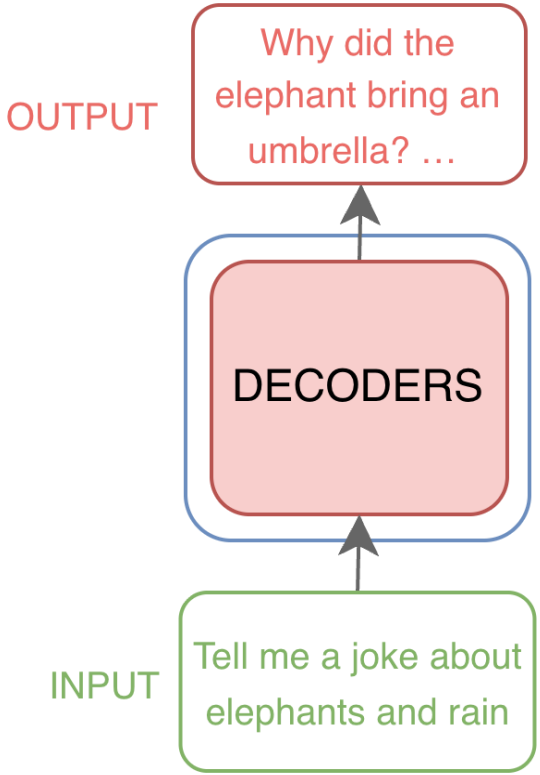
  "Tell me a joke about elephants and rain"

- Output

  Autoregressive next-token prediction conditioned on the input

  "Why did the elephant bring an umbrella? …"

- Expected features
  - Efficient, simple, creative
  - Less controlled; blends prompt and joke



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Evaluation Plan

Humor Quality	Creativity & Diversity	Linguistic Quality	Topic Relevance
- Human Evaluation - Humor Classifier	- n-Novelty - Distinct-n - Self-BLEU-n	- Perplexity - Grammar Checker	- Exact Match - Semantic Similarity

Evaluation Plan

Humor Quality

- Human Evaluation
- Humor Classifier

- Human Evaluation
  - Human judges rate both training and generated jokes
    - Funniness: 1 (not funny) … 5 (very funny)
    - Relevance: 1 (off-topic) … 5 (perfectly on-topic)
    - Coherence: 1 (broken/awkward) … 5 (clear & natural)
    - Originality: 1 (cliché/seen often) … 5 (fresh twist)
    - Safety: OK / Not OK
- Humor Classifier
  - DistilBERT fine-tuned for humor detection, F1 > 0.95
  - Probability of being humorous

Evaluation Plan

Creativity & Diversity

- n-Novelty
- Distinct-n
- Self-BLEU-n

- n-Novelty
  - $\text{n-Novelty}(G, C) = 1 - \frac{|\text{n-grams}(G) \cap \text{n-grams}(C)|}{|\text{n-grams}(C)|}$
  - Higher = less common with corpus → more creative
- Distinct - n
  - $\text{Distinct-}n = \frac{|\text{unique n-grams in G}|}{|\text{all n-grams in G}|}, n \in \{1,2\}$
  - Higher = less repetition → more diverse
- Self-BLEU - n
  - $\text{Self-BLEU-}n = \frac{1}{N}\sum_{i=1}^{N} \text{BLEU}_n\left(G_i, G \backslash \{G_i\}\right), n \in \{2,3\}$
  - Lower = less mode collapse → more unique

Evaluation Plan

Linguistic Quality
- Perplexity - Grammar Checker

- Perplexity
  - Measures how well it reads like real language
  - GPT-2 small, a commonly used benchmark
- Grammar Checker
  - Checks grammatical correctness
  - LanguageTool, open-source on GitHub

Evaluation Plan

Topic Relevance
- Exact Match - Semantic Similarity

- Exact Match
  - verifies the presence of all topic words
- Semantic Similarity
  - captures synonyms or related expressions (e.g., rainy for rain)

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Expected Results

Big Vision	- Model optimization techniques - Competitive performance against LLMs - **Comprehensive quantitative analysis** - Interactive interface
Minimal Goal	- Runnable models - Rated funnier than non-humorous data - Ensure enough linguistic quality - Include ≥ 1 input topic in the joke
Core	- Build 2 different transformer architectures - Complete multi-dimensional evaluation - Fits within our CPU budget

Challenges & Risks

- Limited model capacity
  - A small LM may limit joke quality
  - Overfitting: may memorize phrases
- Evaluation
  - Humor subjectivity: What's "funny" varies by culture and rater.
- Analysis
  - How to reconcile quantitative metrics with human judgments
- Limited compute budget
  - Balance between model size and training efficiency

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Achieved Goals

Total in	: **1,303,811**
Blank after norm	: 17
Too short (<10 chars)	: 150
Exact duplicates	: 537,187
Too long (>1,000 chars)	: 39,670
Final out	: **726,787**

- First 3 goals
- Last Week

1. Data preprocessing:
   - Data collection & merging
   - Data cleaning (table)
   - Topic extraction: POS tagging (spaCy, up to 3 topics)
2. Start coding the Transformer
   - Tokenizer trained and saved (BPE 52K)
   - Started coding decoder:
     - Defined prompt format:
       `"tell me a joke about {topic} [JOKE] {joke}"`
     - Started Decoder class implementation (in progress)
3. Prepared initial talk

References

Datasets

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Tools

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Q & A

Thank you for your attention